

Course Number	0418-475	Course Title	Distributed Database Systems
Credit Hours	3 (3:0:3)	Course Coordinator	Dr. Saud AlDarmi
Compulsory	No	Prerequisite	0418-331, 0418-470

Textbook

<i>Title:</i>	<i>Principles of Distributed Database Systems</i>
<i>Author:</i>	<i>M.T. Oszu, P. Valduriez</i>
<i>Publisher:</i>	<i>3rd edition, Prentice Hall, 2011</i>

References

<i>Title:</i>	<i>Oracle Web Application Programming for PL/SQL Developers</i>
<i>Author:</i>	<i>S. Boardman, M.Caffrey, S. Morese, B. Rosenzweig</i>
<i>Publisher:</i>	<i>Prentice-Hall, Latest Edition</i>

Major Topics Covered in the Course

	Topics	Hours
1	Distributed and centralized databases: Demand for distributed databases, review of relational DBMS, relational algebra, relational calculus, review of computer networks topology, protocol standards.	4
2	Distributed DBMS architecture: Architectural models for distributed DBMS, autonomy, distribution, heterogeneity, distributed database architectures, client/server systems, peer-to-peer systems.	6
3	Distributed database design: Alternative design strategies, horizontal fragmentation, vertical fragmentation, allocation problems, information requirements, solution methods.	8
4	Query Processing: Complexity of relational algebra operations, query processors, types of optimization, query decomposition, data localization, global vs. local query optimization.	6
5	Optimization of distributed queries: Centralized optimization, algorithms for centralized optimization, join vs. semijoin, join ordering in fragment queries, semijoin-based algorithms.	7
6	Transaction management and concurrency control: Transaction management, properties and types of transactions, serializability	8

	theory, distributed concurrency control, locking-based and timestamp-based concurrency control, failures and fault tolerance in distributed databases.	
7	Exams.	3
	Total.	42

Grading

15-20%	Midterm 1
15-20%	Midterm 2
15-20%	Lab
15-0%	Hmk/ quiz
40%	Final

Course Outcomes

Students should be able to:

1. Explain the techniques used for data fragmentation, replication, and allocation during the distributed database design process.
2. Evaluate simple strategies for executing a distributed query to select the strategy that minimizes the amount of data transfer.
3. Explain how the two-phase commit protocol is used to deal with committing a transaction that accesses databases stored on multiple nodes.
4. Describe distributed concurrency control based on the distinguished copy techniques and the voting method.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1	L	M				
2		H				
3	L	H				
4		M				